POSTGRESQL · DBA

# Find the query that's stuck.

One look at pg_stat_activity, then cancel or terminate it safely.

WHY IT MATTERS

# Two kinds of stuck session

A query that never finishes. It keeps a connection busy.

A transaction nobody closed. It keeps its row locks until it ends.

Both show up in one view: pg_stat_activity.

STEP 1

# List what's running

```
SELECT pid,
       now() - query_start AS running_for,
       state,
       wait_event_type,
       left(query, 40) AS query
FROM   pg_stat_activity
WHERE  backend_type = 'client backend'
  AND  state <> 'idle'
  AND  pid <> pg_backend_pid()
ORDER BY running_for DESC;
```

Skips idle connections, background processes and your own session.

REAL OUTPUT

# Two sessions found

| pid | running_for | state | wait |
|---|---|---|---|
| 195 | 00:00:07.99 | active | Timeout |
| 196 | 00:00:07.99 | idle in transaction | Client |

**195** is running a long SELECT.

**196** ran an UPDATE, then never sent COMMIT. It is waiting on the client.

Times are cut to 2 decimals to fit. Tested on PostgreSQL 16.

READ IT RIGHT

# The column that fools people

For an idle in transaction session, **query** is the **last** statement it ran, not something running now.

And **query_start** is when that last statement started.

To see how long the transaction has been open, use xact_start.

STEP 2

# Find open transactions

```
SELECT pid,
       now() - xact_start AS txn_open_for,
       left(query, 40) AS last_query
FROM   pg_stat_activity
WHERE  state = 'idle in transaction';
```

| pid | txn_open_for | last_query |
|---|---|---|
| 196 | 00:00:08.02 | UPDATE orders SET… |

STEP 3 · GENTLE

# pg_cancel_backend

```
SELECT pg_cancel_backend(195);
```

```
 pg_cancel_backend
-------------------
 t
```

What session 195 sees:

```
ERROR:  canceling statement due to user request
```

Stops the **query**. The session stays connected.

STEP 3 · FIRM

# pg_terminate_backend

```
SELECT pg_terminate_backend(196);
```

Ends the whole **session**. Its open transaction is rolled back.

```
SELECT status FROM orders
WHERE  order_id = 42;

 status
--------
 NEW       -- the UPDATE to PAID is gone
```

Allowed for superusers, members of the session's role, or pg_signal_backend. Only superusers can touch superuser sessions.

RECAP · SCREENSHOT THIS

# Stuck session checklist

- ✔ Filter on **state <> 'idle'** and client backends
- ✔ **now() - query_start** for running queries
- ✔ **now() - xact_start** for open transactions
- ✔ Try **pg_cancel_backend** first
- ✔ **pg_terminate_backend** ends it and rolls back

BEFORE YOU GO

# Save this for the next "the DB is slow" call.

Every query and output here was run on PostgreSQL 16.